# Notes: Higgins & Rodriguez (JFE, 2006)

## The Outsourcing of R&D through Acquisitions in the Pharmaceutical Industry

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# 1 Big picture

- **Question.** Why do pharmaceutical firms acquire research-intensive firms, and do these acquisitions successfully outsource research and development?
- **Setting.** The paper studies 160 pharmaceutical and biotechnology acquisitions from 1994 to 2001, a period in which drug patent lives were shortening, product pipelines were becoming more important, and large firms faced pressure to replenish future products.
- **Main motivation.** The authors argue that deteriorating internal R&D productivity motivates firms to acquire external research capabilities. Acquisitions are therefore interpreted as a form of R&D outsourcing.
- **Key measurement contribution.** The paper builds a novel *Desperation Index* to measure whether an acquirer's internal pipeline and product portfolio are deteriorating before an acquisition.
- **Core empirical result 1.** Firms with deteriorating pipelines, especially firms in the most desperate categories, are more likely to undertake an acquisition.
- **Core empirical result 2.** Acquirer announcement returns are positive. The average acquirer cumulative abnormal return is about 3.91%, which is high relative to the typical M&A literature.
- **Core empirical result 3.** Returns are larger when acquirers have prior information about the target, such as prior alliances, pipeline experience, or sales experience in the same therapeutic category.
- **Core empirical result 4.** Post-acquisition real outcomes improve for many acquirers: 71% maintain or improve their product pipeline or portfolio after the acquisition.
- **Economic interpretation.** The paper suggests that acquirers can avoid the winner's curse when they possess information about target R&D assets and when acquisitions are motivated by a real strategic need.
- **Takeaway.** R&D acquisitions in pharmaceuticals can create value when they replenish depleted pipelines and when acquirers have enough pre-acquisition information to value and integrate the target.

# 2 Data and measurement

## 2.1 Acquisition sample

- The sample contains 160 acquisitions related to pharmaceutical research and development from 1994 to 2001.
- Acquisitions are classified into three transaction types:
  - **Type I:** acquisition of biotechnology research and development.
  - **Type II:** acquisition of general research and development.
  - **Type III:** acquisition of a mature product together with research and development capabilities.
- The unit of analysis changes across sections:
  - firm-year observations for acquisition propensity;
  - acquisition announcements for event-study returns;
  - acquisition-level observations for cross-sectional regressions;
  - pre- and post-acquisition firm observations for real post-acquisition outcomes.

**Interpretation:** The sample is deliberately industry-specific. This sacrifices broad external validity but allows the authors to observe detailed pipeline, patent, sales, and alliance variables that are usually unavailable in general M&A samples.

## 2.2 Product exclusivity and sales-weighted patent horizons

- The pharmaceutical industry is highly dependent on patent and regulatory exclusivity.
- The authors use FDA Orange Book and related sources to identify remaining exclusivity years for patented products.
- Fig. 1 shows that aggregate exclusivity years rose sharply in the late 1990s and then began falling around 2000–2001.
- Because not all approved products are equally important, the paper constructs a *sales-weighted exclusivity horizon*.
- A firm's product portfolio is more vulnerable when important products are closer to losing exclusivity.

**Interpretation:** The sales-weighted exclusivity measure captures the condition of the firm's current product portfolio, not merely the number of products under patent protection.

## 2.3 Research pipeline and Score variable

- The authors use NDA pipeline files from 1994 to 2001 to measure potential products in development.
- Pipeline stages include pre-clinical, Phase I, Phase II, Phase III, and FDA-filed/NDA products.
- Clinical probabilities are assigned to each phase to reflect the chance that a potential treatment eventually receives FDA approval.
- The weighted value of a firm's research pipeline is called *Score*.
- A higher Score means that the firm has more products or later-stage products in its pipeline.
- The main object is not only the level of Score but also whether Score is rising or falling around the acquisition.

**Interpretation:** Score is a forward-looking measure of R&D productivity. A falling Score signals that the firm has a weak future product pipeline even if current sales are still healthy.

## 2.4 Desperation Index

The paper combines the trend in Score with the trend in sales-weighted exclusivity to form a four-category Desperation Index:

- **Category I:** increasing or constant Score and increasing or constant weighted sales horizon. This is the least desperate category.
- **Category II:** increasing or constant Score but decreasing weighted sales horizon.
- **Category III:** decreasing Score but increasing or constant weighted sales horizon.
- **Category IV:** decreasing Score and decreasing weighted sales horizon. This is the most desperate category.

The aggregated Desperation Index used in several regressions is

$$Desperation_i = \mathbf{1}\{Category_i \in \{III, IV\}\}. \tag{1}$$

**Interpretation:** Category III and IV firms are experiencing deterioration in the research pipeline. The index is designed to capture the degree to which the firm needs to replenish future products through acquisition.

### 2.5 Alliance and experience variables

- **Pipeline experience:** dummy equal to one if the acquirer has pipeline research in the same therapeutic category as the target.
- **Sales experience:** dummy equal to one if the acquirer has patented pharmaceutical product sales in the same therapeutic category as the target.
- **Alliance:** dummy equal to one if the acquirer has an alliance with the target before the acquisition.
- **Royalty:** dummy equal to one if the acquirer was paying royalties to the target before the acquisition.
- **Early-stage alliance:** dummy equal to one if the acquirer had an early-stage product alliance with the target before the acquisition.

**Interpretation:** These variables proxy for information acquisition. The more the acquirer already knows about the target or therapeutic category, the less likely it is to overpay or buy the wrong asset.

### 2.6 Financial controls and deal characteristics

- **R&D intensity:** R&D expense divided by sales.
- **Free cash flow:** cash-flow measure that can affect acquisition capacity.
- **Tobin's Q:** proxy for investment opportunities.
- **Log market capitalization:** proxy for firm scale.
- **Prior acquisition:** whether the acquirer has acquisition experience in the preceding three years.
- **Stock deal/cash deal:** financing method.
- **Contingent contract:** whether the transaction included contingent payments.
- **Related:** whether acquirer and target operate in the same therapeutic category.
- **Sales force:** whether sales-force personnel are part of the acquisition.
- **International:** whether the acquirer is an international firm.

**Interpretation:** These controls help separate the R&D outsourcing mechanism from general M&A motives, financing effects, and firm scale.

## 3 Models and empirical specifications

### 3.1 Acquisition propensity model

The first empirical question is whether deteriorating R&D productivity makes a firm more likely to acquire an R&D-intensive target. The paper estimates probit models of the form

$$Pr(Acquisition_{i,t} = 1 \mid X_{i,t}) = \Phi(X'_{i,t}\beta), \tag{2}$$

where $Acquisition_{i,t}$ equals one if firm $i$ makes an acquisition in year $t$ and $X_{i,t}$ contains pre-acquisition pipeline, desperation, and financial variables.

The marginal effect is

$$\frac{\partial Pr(Acquisition_{i,t} = 1 \mid X_{i,t})}{\partial x_k} = \phi(X'_{i,t}\beta)\beta_k. \tag{3}$$

**Interpretation:** A positive coefficient on Category III, Category IV, or the aggregated Desperation Index means that deteriorating pipeline conditions increase the probability of acquisition.

### 3.2 Event-study abnormal returns

The paper measures acquirer stock market performance using cumulative abnormal returns:

$$CAR_i(\tau_1, \tau_2) = \sum_{\tau=\tau_1}^{\tau_2} \left(R_{i,\tau} - E[R_{i,\tau}]\right). \tag{4}$$

The main CAR is a three-day event window around the acquisition announcement. The reported average acquirer CAR is 3.91%.

**Interpretation:** The event study asks whether investors value the acquisition at announcement. Positive CARs suggest that the market expects these R&D acquisitions to create value for acquirers.

### 3.3 Univariate CAR tests

The paper compares CARs across deal and information categories:

- related versus unrelated acquisitions;
- stock versus cash financing;
- prior alliances versus no prior alliances;
- prior sales experience versus no prior sales experience;
- prior research/pipeline experience versus no prior experience;
- Type I, Type II, and Type III acquisition subgroups.

**Interpretation:** These tests provide transparent evidence that information advantages and product-market familiarity increase acquisition success.

### 3.4 Cross-sectional CAR regressions

The paper estimates cross-sectional regressions of the form

$$CAR_i = \alpha + \beta_1 PipelineExperience_i + \beta_2 SalesExperience_i + \beta_3 Alliance_i + \beta_4 Desperation_i + \gamma' Controls_i + \varepsilon_i. \tag{5}$$

- *PipelineExperience* and *SalesExperience* proxy for knowledge about the therapeutic category.
- *Alliance* proxies for direct pre-acquisition information about the target.
- *Desperation* captures whether the acquirer is under internal R&D pressure.
- Controls include R&D intensity, prior acquisition activity, deal financing, free cash flow, Tobin's Q, market capitalization, contingent contract, and international status.

**Interpretation:** The main prediction is that information advantages raise CARs, while desperation may lower CARs because desperate acquirers have weaker bargaining positions or more urgent need.

### 3.5 Post-acquisition real-outcome regressions

The paper then studies whether acquisitions improve the acquirer's underlying R&D position. It estimates regressions such as

$$\Delta Score_{i,t+1} = \alpha + \beta Desperation_i + \gamma' Controls_i + u_i, \tag{6}$$

where

$$\Delta Score_{i,t+1} = Score_{i,t+1} - Score_{i,t}. \tag{7}$$

It also estimates regressions for post-acquisition sales changes:

$$\Delta Sales_{i,t+1} = \alpha + \beta Desperation_i + \gamma' Controls_i + u_i. \tag{8}$$

**Interpretation:** These regressions test whether the acquisition actually replenishes the acquirer's pipeline or sales base, not just whether the stock market initially reacts favorably.

## 4 Identification and empirical design

### 4.1 What the paper is trying to identify

- The paper tries to identify whether pharmaceutical acquisitions are used to outsource R&D when internal productivity deteriorates.
- It also tries to identify whether acquirers succeed when they have superior information about the target or therapeutic area.
- The empirical design combines acquisition probability, event-study returns, and post-acquisition product-pipeline outcomes.

**Interpretation:** No single test proves the mechanism. The paper's strength is that several pieces of evidence point in the same direction: desperate firms acquire, informed acquisitions earn positive returns, and many acquirers improve their pipelines afterward.

### 4.2 Why the pharmaceutical setting is useful

- Drug development has observable stages: pre-clinical, Phase I, Phase II, Phase III, and NDA/FDA filing.
- FDA, patent, sales, and alliance data provide objective measures of pipeline health and product-market exposure.
- Acquisitions in this sector often involve intangible assets, making information asymmetry central.
- Pharmaceutical firms face clear product-cycle pressure because patent exclusivity eventually expires.

**Interpretation:** The sector makes it possible to measure both the reason for acquisition and whether the acquisition affected real R&D productivity.

### 4.3 Information asymmetry and winner's curse

- Acquisition targets often possess hard-to-value intangible R&D assets.
- Standard M&A theory predicts that acquirers can overpay if targets have superior information.
- Higgins and Rodriguez argue that prior alliances, sales experience, and pipeline experience reduce this informational disadvantage.
- Better-informed acquirers should earn higher CARs because they are less likely to overbid, buy the wrong target, or fail at integration.

**Interpretation:** The paper's information variables are not merely controls. They are central to the explanation for why acquirers in this sample earn positive returns while acquirers in broad M&A samples often do not.

### 4.4 Why post-acquisition outcomes matter

- Positive event returns could reflect market optimism rather than real improvement.
- The paper therefore examines changes in Score and weighted sales after acquisition.
- The Desperation Index is also recomputed after acquisition to see whether firms move to healthier categories.
- 59% of acquirers improve their desperation category, 12% remain constant, and 29% worsen.

**Interpretation:** Post-acquisition pipeline and sales outcomes are direct measures of whether R&D outsourcing worked.

### 4.5 Limitations

- Acquisitions are not randomly assigned.
- Firms that acquire may differ from non-acquirers in unobserved managerial quality, strategy, or risk appetite.
- Pipeline Score uses clinical probability weights and may not capture all dimensions of drug quality.
- Patent and sales data may miss some private or early-stage R&D projects.
- Announcement returns measure investor expectations, not realized long-run value.
- The sample is limited to 1994–2001 and to biopharmaceutical deals.

**Interpretation:** The paper should be read as strong industry-specific evidence rather than a universal causal estimate for all acquisitions.

# 5 Tables and figures

Visuals are directly extracted from the paper and grouped compactly to save space.

## 5.1 Figure 1 and Figure 2: industry pressure and pipeline trends

**Read:**

- Figure 1 shows the total years of exclusivity remaining for patented products in the FDA Orange Book from 1990 to 2001.
- Exclusivity years rise sharply in the late 1990s and then fall by 2001, reflecting pressure on current product portfolios.
- Figure 2 shows drug candidates by development stage from 1993 to 2001.
- Early-stage candidates increase dramatically, while NDA submissions remain relatively flat.

**Economic message:** Large pharmaceutical firms faced a mismatch: current product exclusivity was under pressure, while many potential products were still far from approval. This creates incentives to acquire external research assets.

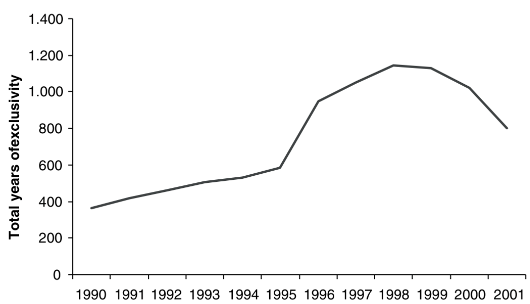


*Figure 1: product exclusivity horizon*

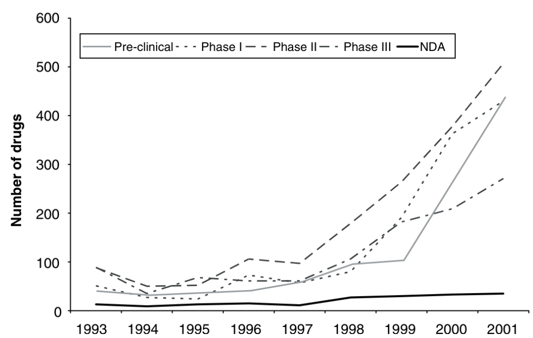


*Figure 2: drug candidates by development stage*

## 5.2 Table 1 and Table 2: variable definitions and descriptive statistics

**Read:**

- Table 1 defines the pipeline, sales, alliance, experience, deal-financing, and financial-control variables used throughout the paper.
- Table 2 shows that the average acquisition sample firm has a Score of 4.49 in the year of acquisition.
- The average pre-acquisition change in Score is 1.39, while the average post-acquisition change is 0.51.
- The average Desperation Index is 0.60, meaning 60% of acquiring firms are in Category III or IV before the acquisition.
- 67% of deals involve a prior alliance and 81% are related acquisitions.

**Economic message:** The sample is heavily concentrated in firms that already face pipeline pressure and in deals where the acquirer has some informational or product-market connection to the target.

Definition and description of regression-model independent variables

Different dependent variables are utilized. Cumulative abnormal returns ($CAR_j$) are used in Table 7 and Δ *Score* and Δ *Sales* are used in additional cross section analyses presented in Table 9.

| Variable | Description |
|---|---|
| *Score* | Weighted value (non monetary) of company research pipeline |
| Δ *Score pre-acquisition* | Pre-acquisition change in weighted value of research pipeline ($score_t - score_{t-1}$) |
| Δ *Score post-acquisition* | Post acquisition change in pipeline weighted value ($score_{t+1} - score_t$) |
| *Log sales level* | Log value of real life-cycle-weighted pharmaceutical sales |
| Δ *Sales pre-acquisition* | Pre-acquisition change in weighted sales ($log\ sales\ level_t - logsaleslevel_{t-1}$) |
| Δ *Sales post-acquisition* | Post acquisition change in weighted sales ($log\ sales\ level_{t+1} - logsaleslevel_t$) |
| *R&D intensity* | Research and development expenses/sales (real 1999 dollars) |
| *Pipeline experience* | Dummy equals 1 if acquiring company has pipeline research in the same therapeutic category as the target firm |
| *Sales experience* | Dummy equals 1 if acquiring company has patented pharmaceutical product sales in the same therapeutic category as the target firm |
| *Alliance* | Dummy equals 1 if acquiring company has alliance with target firm *prior* to the acquisition |
| *Royalty* | Dummy equals 1 if acquiring company was paying a royalty to the target firm prior to the acquisition |
| *Early stage alliance* | Dummy equals 1 if acquiring company has early stage product alliance with target firm prior to acquisition |
| *Desperation Index* (pre-acquisition) | Dummy equals 1 if acquiring company was assigned to desperation Categories III or IV |
| Δ *Desperation Index* post-acquisition | Dummy equals 1if acquiring company improved level of desperation post-acquisition |
| *Prior acquisition* | Dummy equals 1 if acquiring company has prior acquisition experience in the three years before the latest acquisition |
| *Stock deal* | Dummy equals 1 if acquisition was stock financed |
| *Cash deal* | Dummy equals 1 if acquisition was cash financed |
| *Contingent contract* | Dummy equals 1 if contingent contract was present |
| *Related* | Dummy equals 1 if acquiring and target firms operate within the same therapeutic category or categories |
| *Sales force* | Dummy equals 1 if sales force personnel were part of acquisition |
| *Free-cash flow* | Free-cash flow |
| *Tobin* | Tobin's *Q* |
| *Log market cap* | Log of market capitalization |
| *International* | Dummy equals 1 if acquiring firm was an international firm |

*Table 1: variable definitions*

Table 2

Descriptive statistics for firms making 160 research and development related acquisitions in the 1994 to 2001 period

These variables are used in probit results analyzing the characteristics that impact the probability a firm engages in an acquisition (Table 4), analysis of the event study results (Table 5), cross-section results that utilize the cumulative abnormal return as the dependent variable (Table 7), and cross-section results that utilize two separate measures of success as the dependent variable (Table 9).

| Variable | Mean | Standard deviation | Minimum value | Maximum value |
|---|---|---|---|---|
| *Free-cash flow* | 438.03 | 943.77 | −250.87 | 4669.00 |
| *Tobin* | 3.69 | 3.53 | −0.96 | 32.25 |
| *Log market cap* | 7.26 | 2.42 | 2.15 | 12.58 |
| *Score* (in year of acquisition) | 4.49 | 8.66 | 0 | 51.54 |
| Δ *Score pre-acquisition* | 1.39 | 5.06 | −9.95 | 28.15 |
| Δ *Score post-acquisition* | 0.51 | 191.4 | −10.6 | 114.4 |
| *Log sales level* | 3.51 | 5.67 | 0 | 15.58 |
| *Change Sales Level* | 0.30 | 1.72 | −9.71 | 12.15 |
| *R&D intensity* | 3.06 | 26.05 | 0 | 324.98 |
| *Pipeline experience* | 0.36 | 0.48 | 0 | 1 |
| *Sales experience* | 0.23 | 0.42 | 0 | 1 |
| *Alliance* | 0.67 | 0.29 | 0 | 1 |
| *Royalty* | 0.03 | 0.17 | 0 | 1 |
| *Desperation Index* | 0.60 | 0.26 | 0 | 1 |
| *Prior acquisition* | 0.28 | 0.45 | 0 | 1 |
| *Stock deal* | 0.36 | 0.48 | 0 | 1 |
| *Cash deal* | 0.18 | 0.39 | 0 | 1 |
| *Contingent contract* | 0.21 | 0.41 | 0 | 1 |
| *Related* | 0.81 | 0.39 | 0 | 1 |
| *Sales force* | 0.13 | 0.34 | 0 | 1 |
| *International* | 0.24 | 0.43 | 0 | 1 |

*Table 2: descriptive statistics*

## 5.3 Table 3: correlation matrix

**Read:**

- The correlation matrix shows relationships among experience, alliance, desperation, financing, and control variables.
- Pipeline experience and sales experience are positively correlated.
- Desperation Index is positively correlated with pipeline experience and international status, but the correlations are not so high as to imply redundancy.
- Contingent contracts are negatively correlated with Tobin's Q and some information variables.

**Economic message:** The correlation table shows that the information variables are related but distinct. This supports using them separately in the regression specifications.

## 5.4 Table 4: acquisition probability and the Desperation Index

**Read:**

- Table 4 estimates probit models for the probability of making an acquisition.
- In Model 1, pre-acquisition change in Score is negative and significant, implying that firms with improving pipelines are less likely to acquire.
- Category III and IV firms are significantly more likely to acquire than Category I firms.
- Category III has marginal effects around 11–12%, while Category IV has marginal effects around 16%.
- The aggregated Desperation Index is positive and significant in Model 4, with a marginal effect around 13.86%.
- R&D intensity is positive, while log market capitalization is negative and significant in several specifications.

Table 3
Correlation matrix for relevant variables utilized in the following analyses: probit estimation (Table 4); cumulative abnormal return cross-sectional regressions (Table 7); and, post-acquisition cross-sectional regressions (Table 9)

The numbers listed horizontally across the top row correspond to the numbers and variables listed vertically on the table. See Table 1 for variable definitions.

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. Pipeline experience | 1.0000 | | | | | | | | | | | | | | |
| 2. Sales experience | 0.1292 | 1.0000 | | | | | | | | | | | | | |
| 3. Royalty | −0.0669 | −0.0260 | 1.0000 | | | | | | | | | | | | |
| 4. Early stage alliance | −0.0474 | 0.0306 | −0.0007 | 1.0000 | | | | | | | | | | | |
| 5. Equity stake | 0.0428 | 0.1841 | 0.1921 | 0.2491 | 1.0000 | | | | | | | | | | |
| 6. Desperation Index | 0.1373 | 0.3467 | 0.0264 | 0.1888 | 0.1258 | 1.0000 | | | | | | | | | |
| 7. R&D intensity | −0.0494 | −0.0641 | −0.0010 | −0.0580 | −0.0109 | −0.0512 | 1.0000 | | | | | | | | |
| 8. Prior acquisition | 0.1238 | 0.2685 | −0.0484 | 0.0824 | 0.0292 | 0.2744 | −0.0797 | 1.0000 | | | | | | | |
| 9. Stock deal | 0.0861 | 0.0254 | 0.0017 | 0.1438 | 0.1692 | 0.0284 | −0.0821 | −0.0446 | 1.0000 | | | | | | |
| 10. Cash deal | −0.0318 | 0.1644 | 0.1162 | −0.0871 | −0.0896 | 0.0199 | −0.0517 | 0.0574 | −0.3697 | 1.0000 | | | | | |
| 11. Free cash flow | 0.1975 | 0.1786 | 0.0217 | 0.0689 | −0.0064 | 0.2630 | −0.0632 | 0.2848 | −0.0440 | 0.0527 | 1.0000 | | | | |
| 12. Tobin | −0.0077 | −0.0661 | −0.0663 | −0.0511 | −0.0575 | 0.0763 | −0.0224 | 0.0103 | −0.1106 | −0.0490 | 0.1642 | 1.0000 | | | |
| 13. Log market cap | 0.0637 | 0.2251 | −0.0730 | 0.1195 | 0.0531 | 0.4156 | −0.1420 | 0.3667 | −0.0447 | 0.1032 | 0.5913 | 0.1421 | 1.0000 | | |
| 14. Contingent contract | 0.1865 | −0.1283 | −0.0139 | −0.0616 | 0.1009 | −0.1353 | 0.1784 | −0.0415 | 0.0468 | −0.0237 | −0.1558 | −0.1099 | −0.2093 | 1.0000 | |
| 15. International | −0.0318 | 0.2948 | −0.0882 | 0.0079 | 0.0369 | 0.2325 | −0.0563 | 0.1802 | −0.0582 | 0.0431 | −0.0024 | −0.0058 | 0.2003 | −0.2046 | 1.0000 |

Table 4
Probit estimates for our data regress an acquisition indicator (equals 1 if an acquisition took place) on a series of independent variables expected to impact a firm's probability of engaging in an acquisition

The period for this analysis runs from 1994 to 2001. The universe of firms for this analysis includes all firms contained in the Recombinant Capital alliance data set that have pipeline and patented product information available. *Count* is defined as the total number of alliances that the acquiring firm is engaged in on a year-by-year basis. *Desperation Index* Categories III and IV are the most severe levels of firm desperation. The variable *Desperation Index* is an aggregated indicator variable that assumes a value of one if the acquirer is in Category III or IV and is zero otherwise. $\Phi$ is the standard cumulative normal distribution. See Table 1 for all other variable definitions. Firm and time effects were included. We use the White (1980) heteroskedasticity-consistent standard errors to compute the appropriate test statistics included in parentheses below the coefficient estimates. We test the model:

$$P(y_{i,t} \neq 0 \mid x_{i,t}) = \Phi(x_{i,t}^1 + x_{i,t}^2 + x_{i,t}^3 + x_{i,t}^4 + x_{i,t}^5 + x_{i,t}^6 + x_{i,t}^7 + x_{i,t}^8 + x_{i,t}^9 + FE + c),$$
$$\partial\Phi/\partial x_i = \phi(\mathbf{xb})b_i.$$

The dependent variable is *Acquisition Indicator*. *** denotes significance at the 1% level; ** denotes significance at the 5% level; and * denotes significance at the 10% level.

| Independent variable | Model 1 | | Model 2 | | Model 3 | | Model 4 | |
|---|---|---|---|---|---|---|---|---|
| | Model 1 | $\partial\Phi/\partial x$ | Model 2 | $\partial\Phi/\partial x$ | Model 3 | $\partial\Phi/\partial x$ | Model 4 | $\partial\Phi/\partial x$ |
| $x^1$: *Change score* (*pre-acquisition*) | −0.0545 | −0.0111 | | | | | | |
| | (1.91)* | | | | | | | |
| $x^2$: *R&D intensity* | 0.0016 | 0.0003 | 0.0014 | 0.0003 | 0.0016 | 0.0003 | 0.0012 | 0.0002 |
| | (1.84)* | | (1.98)** | | (2.34)** | | (1.89)* | |
| $x^3$: *Log market cap* | −0.0457 | | −0.0965 | −0.0214 | −0.0881 | −0.0196 | −0.0894 | −0.0201 |
| | (1.09) | | (2.77)*** | | (2.83)*** | | (2.62)*** | |
| $x^4$: *Count* | 0.0086 | | 0.0033 | | | | 0.0019 | |
| | (0.80) | | (0.37) | | | | (0.22) | |
| $x^5$: *Desperation Index* (*Category* I) | | | | | | | | |
| $x^6$: *Desperation Index* (*Category* II) | | | −0.4640 | | −0.4497 | | | |
| | | | (1.44) | | (1.39) | | | |
| $x^7$: *Desperation Index* (*Category* III) | | | 0.4416 | 0.1141 | 0.4537 | 0.1177 | | |
| | | | (2.14)** | | (2.24)** | | | |
| $x^8$: *Desperation Index* (*Category* IV) | | | 0.5668 | 0.1601 | 0.5776 | 0.1638 | | |
| | | | (1.83)* | | (1.86)* | | | |
| $x^9$: *Desperation Index* | | | | | | | 0.5314 | 0.1386 |
| | | | | | | | (2.96)*** | |
| *c*: *Constant* | −0.8285 | | −0.3479 | | −0.3874 | | −0.3867 | |
| | (2.02)** | | (1.08) | | (1.21) | | (1.21) | |
| Fixed effects | Yes | | Yes | | Yes | | Yes | |
| $N$ | 424 | | 424 | | 424 | | 424 | |
| Pseudo $R^2$ | 0.04 | | 0.04 | | 0.05 | | 0.05 | |
| Log likelihood | −122.96 | | −172.09 | | −172.17 | | −173.23 | |

**Economic message:** Firms with deteriorating pipelines are more likely to acquire. This supports the paper's central claim that acquisitions are used to outsource or replenish R&D when internal productivity weakens.

## 5.5 Table 5 and Table 6: event-study returns and dispersion

**Read:**

- Table 5 reports an average acquirer CAR of 3.91%, significant at the 1% level.
- Type I, II, and III transactions all have positive CARs: 2.81%, 4.29%, and 5.29%, respectively.
- Related acquisitions have average CARs of 3.51% overall.
- Deals with prior alliances have higher returns than those with no prior alliance.
- Deals with prior sales experience have especially high CARs, 6.99% overall.
- Deals with prior research experience also earn higher CARs, 5.08% overall.
- Table 6 shows that only four CARs are less than -15%, while sixteen are greater than 15%.

**Economic message:** The event-study evidence shows that these acquisitions create value for acquirers on average. The higher returns for prior alliance, sales, and research experience indicate that information advantages matter.

Table 5
Event study estimates evaluating the cumulative abnormal return versus the presence of various independent variables

Type I transactions involve the acquisition of biotechnology research and development; Type II transactions involve the acquisition of general research and development; and Type III transactions involve the acquisition of a mature product along with research and development. *T*-test measures whether each of the subcategories (for the overall sample) is statistically different from each other (difference of the means). *t*-statistics are reported in parentheses. See Table 1 for definitions of variables. *** denotes significance at the 1% level; ** denotes significance at the 5% level; and * denotes significance at the 10% level.

| | Overall | *t*-test | Type I | Type II | Type III |
|---|---|---|---|---|---|
| *Acquirer CAR* | 3.91% (4.95)*** $N = 160$ | | 2.81% (2.61)*** $N = 91$ | 4.29% (2.41)** $N = 27$ | 5.29% (3.27)*** $N = 42$ |
| *Relatedness* | 3.51% (4.24)*** $N = 130$ | | 2.49% (2.25)** $N = 68$ | 4.00% (2.01)** $N = 24$ | 4.32% (2.88)*** $N = 36$ |
| *Financing* | | 2.19** | | | |
| *Stock payment* | 3.87% (2.92)*** $N = 58$ | | 3.22% (2.13)** $N = 35$ | 3.36% (0.68) $N = 8$ | 3.71% (1.67)*** $N = 14$ |
| *Cash payment* | 2.38% (2.47)** $N = 29$ | | 2.35% (1.36) $N = 13$ | 2.24% (1.71)* $N = 7$ | 4.08% (1.47) $N = 9$ |
| *Alliances* | | 2.58*** | | | |
| *Prior alliances* | 4.30% (4.14)*** $N = 94$ | | 4.58% (3.25)*** $N = 54$ | 2.11% (1.39) $N = 18$ | 4.22% (1.82) $N = 21$ |
| *No prior alliances* | 3.36% (2.73)*** $N = 66$ | | 0.17% (0.13) $N = 35$ | 8.65% (2.07)** $N = 9$ | 6.37% (2.77)*** $N = 21$ |
| *Sales experience* | | 2.12** | | | |
| *Prior sales* | 6.99% (3.29)*** $N = 128$ | | 4.28% (2.19)** $N = 75$ | 4.50% (1.49) $N = 17$ | 10.56% (1.91)* $N = 31$ |
| *No prior sales* | 3.08% (4.85)*** $N = 34$ | | 2.58% (2.46)** $N = 14$ | 4.17% (4.36)*** $N = 10$ | 3.43% (3.32)*** $N = 11$ |
| *Research experience* | | 2.10*** | | | |
| *Prior experience* | 5.08% (3.71)*** $N = 103$ | | 2.40% (2.46)** $N = 61$ | 9.08% (4.36)*** $N = 15$ | 12.35% (3.32)*** $N = 25$ |
| *No prior experience* | 3.24% | | 3.03% | 6.95% | 0.50% |

*Table 5: univariate CARs*

able 6
he dispersion of cumulative abnormal returns (*CAR*s) are presented

Following Dodd and Warner (1983), standardized abnormal returns (*SAR*s) are computed by dividing the bnormal return (AR) by its standard deviation. These standardized abnormal returns (*SAR*s) are then ggregated over the number of the days in the event window, *k*, to generate a cumulative abnormal return.

| lagnitude | Number of observed abnormal returns |
|---|---|
| $AR \leqslant -15.0\%$ | 4 |
| $15.0\% < CAR < -10.0\%$ | 3 |
| $10.0\% \leqslant CAR < -5.0\%$ | 12 |
| $5.0\% \leqslant CAR < 0.0\%$ | 41 |
| $0\% \leqslant CAR < 5.0\%$ | 38 |
| $0\% \leqslant CAR < 10.0\%$ | 29 |
| $0.0\% \leqslant CAR < 15.0\%$ | 17 |
| $5.0\% \leqslant CAR$ | 16 |

*Table 6: CAR dispersion*

## 5.6 Table 7: cross-sectional regressions of acquirer CARs

**Read:**

- Table 7 regresses acquirer CARs on information, desperation, financial, and deal variables.
- Pipeline experience is positive, with coefficients around 0.018–0.025.
- Sales experience is positive and highly significant in all six models, with coefficients around 0.041–0.057.
- Alliance is positive and highly significant in all included specifications, around 0.032–0.040.
- Desperation Index is negative and statistically significant, with coefficients around -0.036 to -0.044.
- Prior acquisition is negative and significant in several models.
- Log market capitalization is negative in all specifications and significant in Model 6.

**Economic message:** Investors reward acquisitions when the acquirer has information about the target or therapeutic area. But they discount acquisitions by desperate acquirers, consistent with weaker bargaining position or urgency.

## 5.7 Table 8: changes in Desperation Index and post-acquisition pipeline outcomes

**Read:**

- Panel A reports the pre- and post-acquisition distribution of firms across Desperation Index categories.
- Before acquisition, 45% of firms are in Category IV, the most desperate category.
- After acquisition, only 11.88% are in Category IV, while Category I increases from 10.63% to 22.50%.
- The paper reports that 94 firms, or 59%, improve their desperation level; 19 firms, or 12%, remain constant; and 47 firms, or 29%, worsen.

Table 7

Cross-sectional regression estimates and independent variables from regressing the cumulative abnormal return (*CAR*) on selected independent variables for 160 announcements of acquisitions relating to the outsourcing of research and development

*CAR*s are from the three-day event window. The period for this analysis runs from 1994 to 2001. Year fixed effects are included in all specifications. The White (1980) heteroskedasticity-consistent t-statistics are reported in parentheses. See Table 1 for variable definitions. *** denotes significance at the 1% level; ** denotes significance at the 5% level; and * denotes significance at the 10% level.

The dependent variable is *Acquirer CAR*

| Independent variable | Model 1 | Model 2 | Model 3 | Model 4 | Model 5 | Model 6 |
|---|---|---|---|---|---|---|
| *Pipeline experience* | 0.0249 | 0.0227 | 0.0184 | 0.0211 | | |
| | (1.73)* | (1.11) | (1.24) | (1.14) | | |
| *Sales experience* | 0.0565 | 0.0527 | 0.0544 | 0.0529 | 0.0557 | 0.0408 |
| | (3.01)*** | (2.89)*** | (3.12)*** | (2.93)*** | (3.12)*** | (2.65)*** |
| *Alliance* | 0.0348 | 0.0374 | 0.0396 | 0.0347 | 0.0321 | 0.0355 |
| | (3.21)*** | (3.28)*** | (3.11)*** | (3.59)*** | (2.99)*** | (2.67)*** |
| *Desperation Index* | −0.0442 | −0.0428 | −0.0436 | −0.0406 | −0.0362 | |
| | (2.41)** | (2.34)** | (2.42)** | (2.24)** | (2.26)** | |
| *R&D intensity* | 0.0003 | | 0.0004 | | | |
| | (0.91) | | (0.95) | | | |
| *Prior acquisition* | −0.0211 | −0.0204 | −0.0185 | −0.0204 | −0.0174 | −0.0219 |
| | (2.34)** | (2.31)** | (2.17)** | (2.31)** | (2.12)** | (2.41)** |
| *Stock deal* | −0.0117 | −0.0069 | | | | |
| | (0.65) | (0.42) | | | | |
| *Cash deal* | −0.0202 | | | | | |
| | (0.99) | | | | | |
| *Free-cash flow* | −0.0005 | −0.0041 | | −0.0040 | | |
| | (0.41) | (0.39) | | (0.45) | | |
| *Tobin* | 0.0003 | 0.0002 | | 0.0002 | | −0.0002 |
| | (1.14) | (0.11) | | (0.11) | | (0.11) |
| *Log market cap* | −0.0024 | −0.0031 | −0.0035 | −0.0028 | −0.0041 | −0.0057 |
| | (0.45) | (0.62) | (0.98) | (0.56) | (1.22) | (1.73)* |
| *Contingent contract* | 0.0036 | | | | | |
| | (0.16) | | | | | |
| *International* | 0.0108 | 0.0085 | 0.0159 | | | |
| | (0.66) | (0.54) | (0.75) | | | |
| *Constant* | 0.0576 | 0.0591 | 0.0528 | 0.0535 | 0.0711 | 0.0816 |
| | (1.39) | (1.58) | (1.77)* | (1.38) | (2.43)** | (2.86)*** |
| *N* | 155 | 155 | 160 | 155 | 160 | 155 |
| $R^2$ | 0.26 | 0.25 | 0.22 | 0.24 | 0.22 | 0.22 |
| *F*-statistic | 2.78 | 2.64 | 2.91 | 2.87 | 3.12 | 2.58 |

Table 8

*Panel A: distribution of acquiring firms across desperation categories*

Changes in *Desperation Index* pre- and post-acquisition. Classifications are as follows: Category I indicates firms with increasing or constant score values and increasing weighted sales; Category II indicates firms with increasing or constant score values and decreasing weighted sales; Category III indicates firms with decreasing score values and increasing or constant weighted sales; and, Category IV indicates firms with decreasing score values and decreasing weighted sales. Category I firms are less desperate than those firms in Category IV. 94 firms (59%) improved their level of desperation. 19 firms (12%) had their level of desperation remain constant. 47 firms (29%) had a worsening of their level of desperation.

| Pre-acquisition | | | Post-acquisition | | |
|---|---|---|---|---|---|
| Category | Frequency | Percent (%) | Category | Frequency | Percent (%) |
| Category I (least desperate) | 17 | 10.63 | Category I | 36 | 22.50 |
| Category II | 47 | 29.38 | Category II | 72 | 45.00 |
| Category III | 24 | 15.00 | Category III | 33 | 20.63 |
| Category IV (most desperate) | 72 | 45.00 | Category IV | 19 | 11.88 |

*Panel B: distribution of acquiring firms across various post-acquisition measures*

Changes in sales and pipeline score values in the calendar year following the acquisition. The *Score* value is a proxy for the overall health of a firm's research pipeline. An increase in score value would signal that the firm's pipeline has strengthened. Δ *Score* is defined as $score_{t+1} - score_t$,. Δ *Sales* is defined as $sales_{t+1} - sales_t$. Classifications are as follows: Category I indicates firms with increasing or constant score values and increasing weighted sales; Category II indicates firms with increasing or constant score values and decreasing weighted sales; Category III indicates firms with decreasing score values and increasing or constant weighted sales; and Category IV indicates firms with decreasing score values and decreasing weighted sales. Weighted sales values are log values of 1999 constant dollars.

| Variable and category | Number of firms | Mean | Standard Deviation | Minimum | Maximum |
|---|---|---|---|---|---|
| Δ *Score* | | | | | |
| Category I | 36 | 116.91 | 167.14 | 0 | 636.2 |
| Category II | 72 | 40.83 | 142.34 | 0 | 1143.8 |
| Category III | 33 | −173.10 | 228.06 | −1059.6 | −3 |
| Category IV | 19 | −71.34 | 66.07 | −273.6 | −3 |
| Δ *Sales* | | | | | |
| Category I | 36 | 1.0966 | 2.4109 | 0.2436 | 10.82 |
| Category II | 72 | −0.2784 | 1.4465 | −9.7067 | −0.0067 |
| Category III | 33 | 0.4872 | 1.5877 | 0 | 6.5853 |
| Category IV | 19 | −0.8373 | 3.2896 | −14.345 | −1.5194 |

- Panel B reports changes in Score and weighted sales across categories.
- Category I firms have an average Score increase of 116.91, while Category IV firms have an average Score decrease of -71.34.
- Weighted sales changes are positive for Category I and III and negative for Category II and IV.

**Economic message:** Many acquisitions improve acquirers' real product pipeline or sales positions. This post-acquisition evidence supports the view that R&D outsourcing through acquisition can work.

## 5.8 Table 9: post-acquisition Score and sales regressions

**Read:**

- Table 9 studies two post-acquisition outcomes: $\Delta Score$ and $\Delta Sales$.
- In the $\Delta Score$ regressions, Desperation Index is positive and significant in several specifications.
- Desperate firms therefore tend to experience positive post-acquisition changes in pipeline Score.
- In the $\Delta Sales$ regressions, pipeline experience is positive and significant in several specifications.
- Sales force is positive and significant in the sales regressions, consistent with sales assets being important for product commercialization.
- The regressions suggest that acquisition success depends on matching target assets to the acquirer's capabilities and needs.

**Economic message:** The acquisition does not only produce an announcement return. For many acquirers, it improves the measurable pipeline or product-sales position after the acquisition.

Table 9
Cross-sectional regression estimates and independent variables from two separate regressions are reported

First, Δ *Score*, defined as $score_{t+1} - score_t$, is regressed on a series of independent variables thought to have an impact on the firm's underlying research portfolio. Second, Δ *Sales*, defined as $sales_{t+1} - sales_t$, is regressed on a series of independent variables thought to have an impact on the firm's underlying weighted sales. The period for this analysis runs from 1994 to 2001. The White (1980) heteroskedasticity-consistent *t*-statistics are reported in parentheses. See Table 1 for variable definitions. *** denotes significance at the 1% level; ** denotes significance at the 5% level; and * denotes significance at the 10% level.

| Dependent variable | Independent variables | Mode 1 | Model 2 | Model 3 | Model 4 | Model 5 |
|---|---|---|---|---|---|---|
| Δ *Score post-acquisition* | *R&D intensity* | −0.0357 | −0.1566 | −0.1910 | −0.2131 | −0.0041 |
| | | (0.35) | (1.63) | (2.02)$^{**}$ | (2.38)$^{**}$ | (0.40) |
| | *Pipeline experience* | 40.883 | 32.041 | 38.200 | | |
| | | (1.41) | (1.09) | (1.23) | | |
| | *Sales experience* | 57.169 | 73.316 | | | |
| | | (1.14) | (1.44) | | | |
| | *Desperation Index* | 68.592 | 58.326 | 64.637 | 64.857 | 77.116 |
| | | (2.10)$^{**}$ | (1.80)$^{*}$ | (1.94)$^{*}$ | (1.94)$^{*}$ | (2.42)$^{**}$ |
| | *Free-cash flow* | −0.0641 | −0.0373 | −0.0301 | −0.0262 | −0.0574 |
| | | (1.65)$^{*}$ | (1.10) | (0.84) | (0.71) | (1.40) |
| | *Tobin* | −2.476 | | −2.104 | −2.339 | |
| | | (0.73) | | (0.83) | (0.98) | |
| | *Log market cap* | 19.784 | | | | 20.356 |
| | | (2.46)$^{**}$ | | | | (2.51)$^{**}$ |
| | *Constant* | −176.25 | −46.119 | −30.651 | −18.081 | −169.80 |
| | | (3.14)$^{***}$ | (1.70)$^{*}$ | (0.96) | (0.64) | (3.06)$^{*}$ |
| *N* | | 155 | 155 | 155 | 155 | 155 |
| $R^2$ | | 0.12 | 0.08 | 0.06 | 0.05 | 0.09 |
| *F*-statistic | | 13.41 | 17.76 | 16.10 | 20.37 | 23.02 |
| Δ *Sales post-acquisition* | *R&D intensity* | 0.0011 | | | | |
| | | (1.08) | | | | |
| | *Pipeline experience* | 0.7572 | 0.7596 | 0.7257 | 0.5264 | 0.5098 |
| | | (2.18)$^{**}$ | (2.19)$^{**}$ | (2.10)$^{**}$ | (1.32) | (1.32) |
| | *Sales experience* | −0.2547 | −0.2303 | | | |
| | | (0.57) | (0.52) | | | |
| | *Desperation Index* | 0.6012 | 0.6143 | 0.5713 | 0.5742 | 0.5841 |
| | | (1.67)$^{*}$ | (1.71)$^{*}$ | (1.78)$^{*}$ | (1.73)$^{*}$ | (1.80)$^{*}$ |
| | *Free-cash flow* | −0.0059 | −0.0006 | −0.0005 | | |
| | | (1.43) | (1.51) | (1.59) | | |
| | *Tobin* | −0.0322 | | | −0.0435 | |
| | | (0.72) | | | (1.06) | |
| | *Log market cap* | −0.0456 | −0.0397 | | −0.1003 | −0.1041 |
| | | (1.32) | (0.35) | | (1.32) | (1.42) |
| | *Sales force* | 1.4304 | 1.3800 | 1.3027 | 1.4068 | 1.3618 |
| | | (2.11)$^{**}$ | (2.11)$^{**}$ | (2.13)$^{**}$ | (2.10)$^{**}$ | (2.09)$^{**}$ |
| | *Constant* | −0.5992 | −0.6622 | −0.3994 | −0.3017 | −0.1675 |
| | | (1.24) | (1.41) | (1.36) | (0.60) | (0.36) |
| *N* | | 155 | 155 | 155 | 155 | 160 |
| $R^2$ | | 0.15 | 0.15 | 0.15 | 0.11 | 0.10 |
| *F*-statistic | | 1.74 | 1.86 | 2.31 | 1.65 | 2.04 |

# 6 Short summary

- The paper studies 160 pharmaceutical R&D acquisitions from 1994 to 2001.
- It argues that firms use acquisitions to outsource R&D when internal pipelines deteriorate.
- The paper constructs a Desperation Index from changes in pipeline Score and sales-weighted product exclusivity.
- Desperate firms, especially Category III and IV firms, are more likely to acquire.
- Acquirers earn positive announcement returns averaging 3.91%.
- Returns are higher when acquirers have prior alliances, sales experience, or pipeline experience related to the target.
- Cross-sectional regressions show that sales experience and alliances strongly raise acquirer CARs.
- Desperation reduces CARs, suggesting that urgent need weakens bargaining power or raises integration risk.
- Post-acquisition analysis shows that many acquirers improve their pipeline or product portfolio after acquisition.

- Overall, R&D acquisitions can create value when they combine strategic need with information advantages.

# 7 Conclusion

## 7.1 Core conclusion

Higgins and Rodriguez show that pharmaceutical firms use acquisitions as a way to outsource R&D and replenish deteriorating pipelines. The central evidence is that firms with worsening internal R&D productivity are more likely to acquire, and that many acquisitions improve real pipeline outcomes.

## 7.2 Economic intuition

The economic intuition is based on information and strategic need. A firm facing declining pipeline health has a strong reason to seek external research assets. But R&D assets are hard to value. Acquirers avoid overpaying when they already know the target or therapeutic area through alliances, pipeline experience, or sales experience. These information advantages make the acquisition less subject to the winner's curse.

## 7.3 Takeaway

The paper shows that acquisition performance depends on the reason for the acquisition and the acquirer's information set. Broad M&A studies often find weak or negative acquirer returns, but in this specific setting, acquisitions can create value because they solve a measurable R&D pipeline problem and because experienced acquirers know what they are buying.